Siva Nagendra Savarapu

Senior Pipeline Engineer · Generative AI & 3D · OpenUSD · Unreal Engine · Python · C++

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SUMMARY

Senior Pipeline Engineer and Generative AI specialist with 10+ years delivering scalable production pipeline systems for top-tier VFX and game studios including Epic Games, Sony Pictures Imageworks, Animal Logic, and MPC. Expert in OpenUSD, Python/C++, and DCC integration across Unreal Engine, Maya, and Houdini. At Epic Games, among the first engineers to deploy Model Context Protocol (MCP) in a major game studio, contributed to the GenStudio generative AI product, and independently built an agentic C++ UE Runtime system combining Claude, Gemini, Depth Pro, Grounding DINO, SAM2, and mesh generation models into a single prompt-driven 3D scene construction pipeline. Seeking Senior Pipeline Engineer, ML/AI Engineer, or Technical AI roles at the intersection of generative AI and 3D/VFX/game production.

EXPERIENCE

Senior Pipeline Engineer - Epic Games

May 2025 - Mar 2026 · Vancouver, Canada

- Worked on a team building the **Fortnite** asset pipeline spanning Modelling, Texturing, Rigging, and Animation, engineering Python and C++ tools across Maya and Unreal Engine including a Rig Browser via OpenAssetIO, Maya to UE animation import/export, and ShotGrid production tracking automation to accelerate asset creation and eliminate manual handoff overhead across all departments.
- Designed and deployed **Model Context Protocol (MCP) servers for Maya and Unreal Engine**, integrating Claude Code into Epic's live production pipeline, among the earliest production MCP deployments at a major game studio.
- Engineered a **C++ Unreal Engine Runtime prototype** combining Claude (Anthropic Agent SDK) with UE MCP, Gemini/Meshy/Tripo for mesh generation, Depth Pro for depth estimation, Grounding DINO for object detection, and SAM2 for segmentation, delivering an end-to-end agentic 3D scene construction system from a single user prompt or image with mesh placement via a custom raycasting algorithm and GLTF plugin.
- Contributed to **GenStudio**, an internal generative AI product integrating Gemini, OpenAI image models, Meshy, Tripo, and Rodin mesh generation models for AI-assisted 3D asset creation workflows.
- Built a **RAG-powered pipeline assistant** that auto-generates visual workflow graphs (analogous to UE Blueprints) from natural language queries, with a custom RAG backend built on internal pipeline documentation.
- Implemented **C2PA content provenance workflows** across GenStudio, ensuring cryptographic content credential tracking for all AI-generated image outputs in compliance with emerging content authenticity standards.

Senior Software Engineer - Sony Pictures Imageworks

Jul 2023 - May 2025 · Vancouver, Canada

- Engineered an **OpenUSD and Alembic-based asset transfer pipeline from Houdini to Unreal Engine**, preserving textures, materials, and animation data through the full handoff, significantly accelerating the real-time layout process by eliminating manual re-assembly of assets inside UE.
- Built and maintained OpenUSD pipelines for VFX and Animation productions using Python, C++, and PySide6, covering asset and shot workflows across Modelling, Layout, and Animation departments.
- Developed DCC integrations across Maya, Houdini, and Unreal Engine ensuring pipeline data interoperability and compatibility across departments and studios.
- Designed modular show-specific tools in PySide6 and web frameworks deployed inside Maya and Unreal Engine based on production requirements.
- Contributed to productions including **Fantastic Four: First Steps** and **KPop Demon Hunters**.

Core Assets Pipeline Technical Director - Animal Logic
Aug 2021 - May 2023 · Sydney, Australia

- Developed and maintained core Asset Pipeline systems using Pixar's USD, Houdini, Maya, and Python, including USD schema extensions and pipeline plugin development.
- Designed UsdShade material workflows to improve USD data read/write fidelity; improved USD Assembly pipeline for environment workflows and asset validation systems.
- Contributed heavily to USD texture/material workflows for the **ALab public dataset release**.
- Enhanced Houdini UX for artists via Python and PySide6 tooling, reducing friction in day-to-day pipeline interactions.

Technical Director - MPC Film
Nov 2018 - Jul 2021 · Bangalore, India

- Built a Real-Time Asset presentation pipeline in Unreal Engine and developed MPC's core production pipeline tools using Python and PySide6, supporting artists across Rigging, Groom, Lighting, and FX departments.
- Built ION containers to manage reproducible software environments across shows.
- Contributed to productions including **WandaVision** and **Top Gun: Maverick**.

Texture & Lookdev Artist - Rhythm & Hues / Prana Studios
Sep 2015 - Nov 2018 · Mumbai, India

- Worked on **Hellboy (2019)**, **2.0 (2019)**, Fear the Walking Dead, and additional productions.
- Developed Lookdev and Lighting workflows with ImageEngine's Gaffer; trained the team in Gaffer, Substance Painter, and Mari.
- Worked on Unreal Engine and Unity for Previz and real-time VR projects.

EDUCATION

Bachelor's Degree, Computer Science - Acharya Nagarjuna University, India · 2013 - 2016

SKILLS

Languages: Python, C++, MEL, JavaScript

AI/ML: LLM Integration, RAG, MCP, Stable Diffusion, Grounding DINO, Depth Pro, SAM2, VAE, Diffusion Models, C2PA, PyTorch

UI Frameworks: PySide6/Qt6, web frameworks

Tools: Git, GitHub, Docker, GLTF

AI APIs & SDKs: Anthropic Agent SDK, OpenAI API, Gemini API, Meshy, Tripo, Rodin

Pipeline & DCCs: OpenUSD, OpenAssetIO, ShotGrid, Maya, Houdini, Unreal Engine, Nuke, Gaffer, Alembic

Rendering & Real-Time: UE5, PBR workflows, Arnold, Renderman, Substance Painter/Designer, Virtual Production